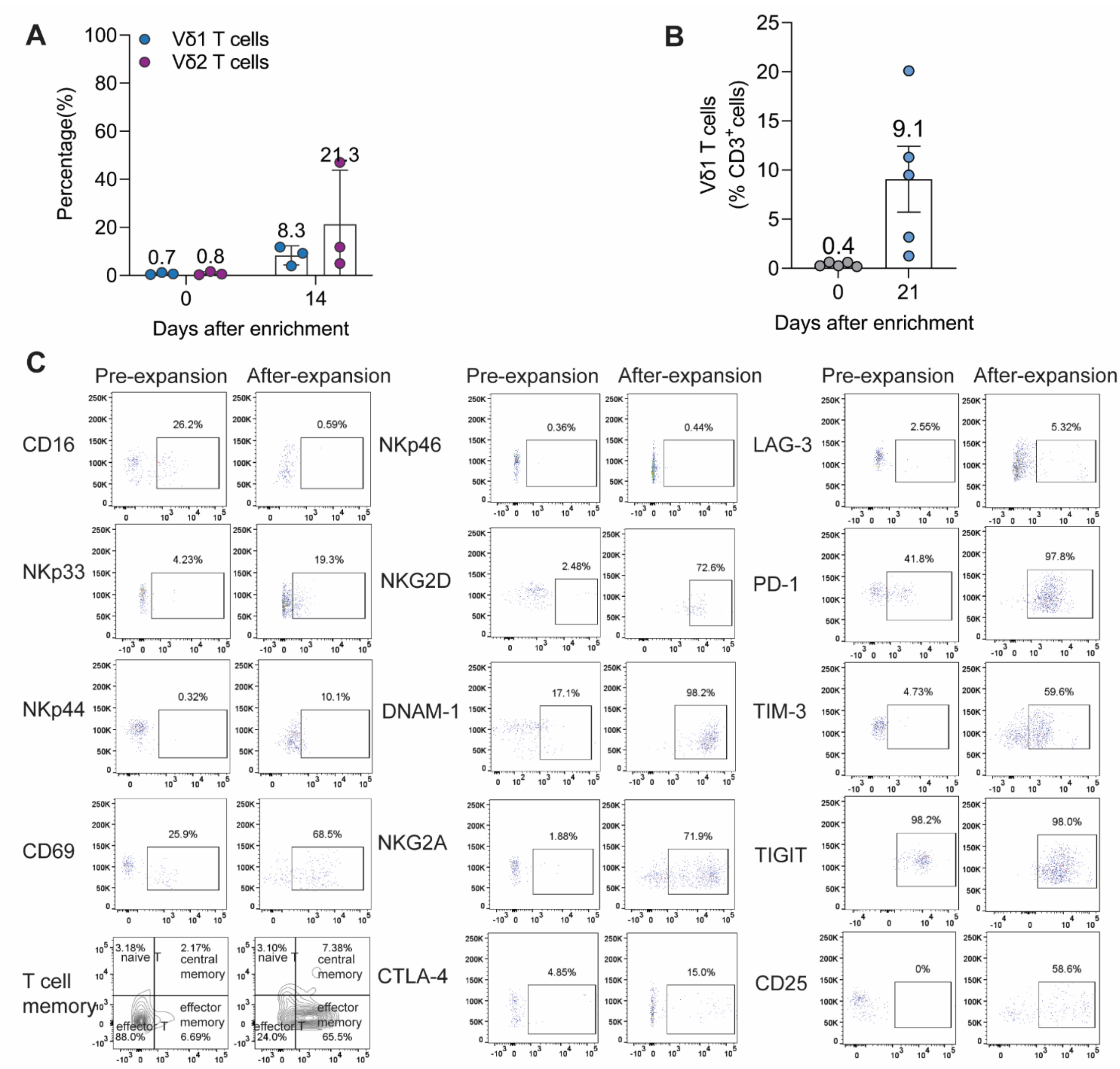
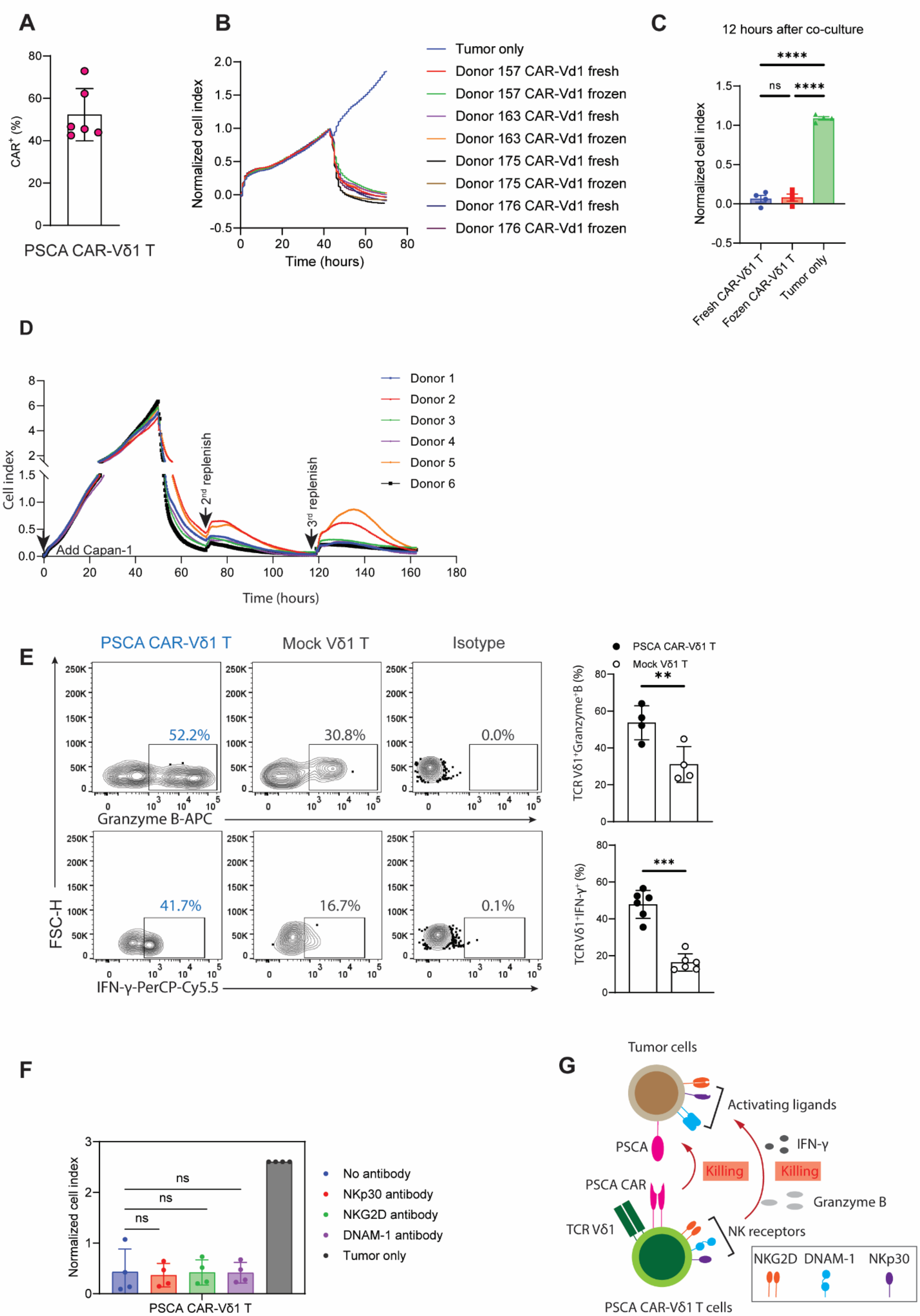

A
Vδ1 T cells
Vδ2 T cells
Percentage(%)
0.7
0.8
8.3
21.3
Days after enrichment
B
Vδ1 T cells (% CD3+cells)
0.4
9.1
Days after enrichment
C
Pre-expansion
After-expansion
CD16
26.2%
0.59%
NKp33
4.23%
19.3%
NKp44
0.32%
10.1%
CD69
25.9%
68.5%
T cell memory
3.18% naive T
2.17% central memory
effector T 88.0%
effector memory 6.69%
3.10% naive T
7.38% central memory
effector T 24.0%
effector memory 65.5%
NKp46
0.36%
0.44%
NKG2D
2.48%
72.6%
DNAM-1
17.1%
98.2%
NKG2A
1.88%
71.9%
CTLA-4
4.85%
15.0%
LAG-3
2.55%
5.32%
PD-1
41.8%
97.8%
TIM-3
4.73%
59.6%
TIGIT
98.2%
98.0%
CD25
0%
58.6%

Supplementary Figure 1. γδ T cells were enriched and expanded from peripheral blood (related to Figure 1).

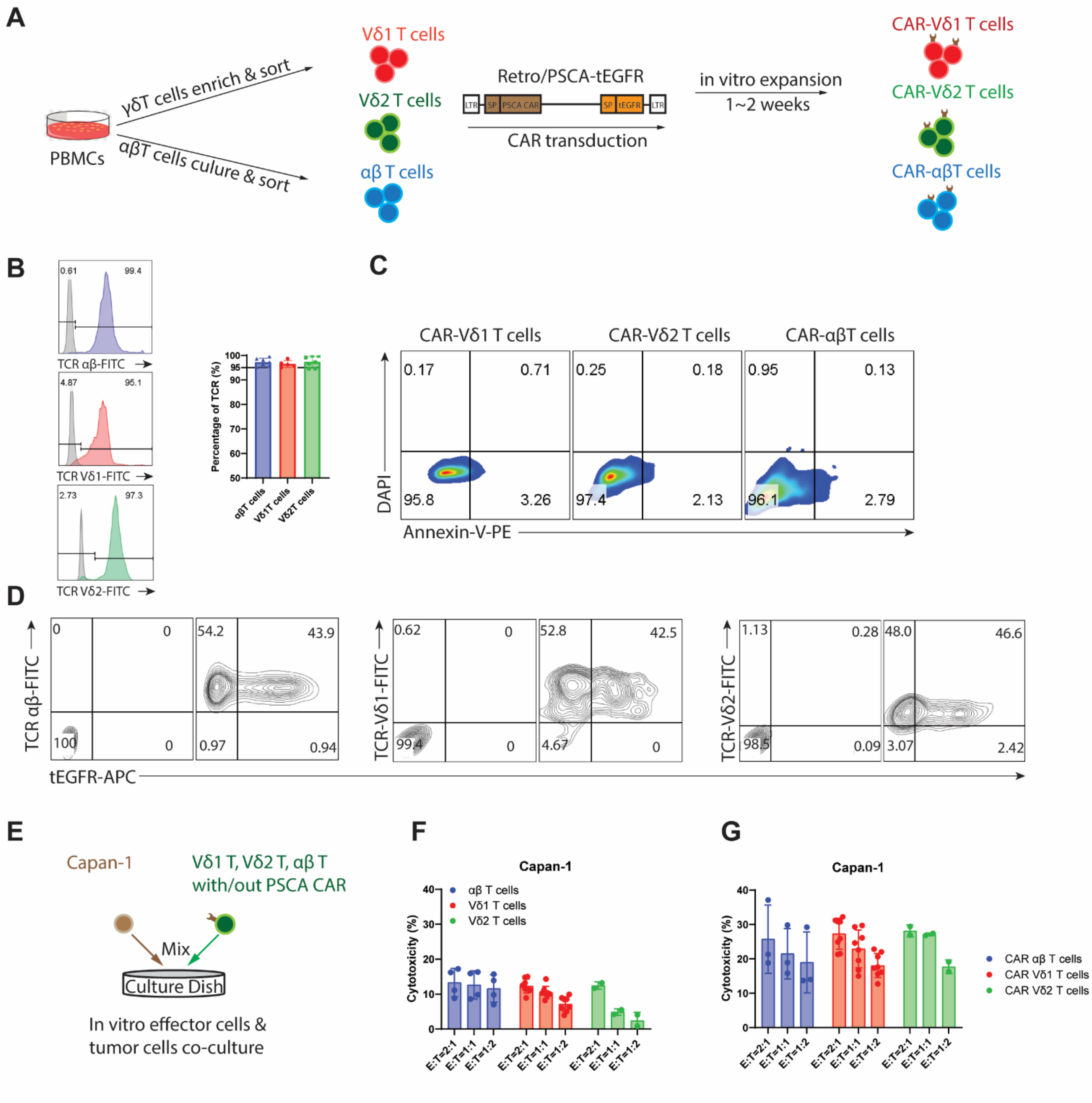
(A) Percentage of Vδ1 and Vδ2 T cells after *in vitro* enrichment. Mean values are indicated on top of each column. **(B)** Vδ1 T cells percentage after 21 days *in vitro* enrichment. Mean values are indicated on top of each column. **(C)** Representative flow plots of Vδ1 T cells before and after expansion.



A
CAR+ (%)
PSCA CAR-Vδ1 T
B
Normalized cell index
Time (hours)
Tumor only
Donor 157 CAR-Vd1 fresh
Donor 157 CAR-Vd1 frozen
Donor 163 CAR-Vd1 fresh
Donor 163 CAR-Vd1 frozen
Donor 175 CAR-Vd1 fresh
Donor 175 CAR-Vd1 frozen
Donor 176 CAR-Vd1 fresh
Donor 176 CAR-Vd1 frozen
C
12 hours after co-culture
Normalized cell index
****
ns
****
Fresh CAR-Vδ1 T
Fozen CAR-Vδ1 T
Tumor only
D
Cell index
Add Capan-1
2nd replenish
3rd replenish
Time (hours)
Donor 1
Donor 2
Donor 3
Donor 4
Donor 5
Donor 6
E
PSCA CAR-Vδ1 T
Mock Vδ1 T
Isotype
52.2%
30.8%
0.0%
Granzyme B-APC
41.7%
16.7%
0.1%
FSC-H
IFN-γ-PerCP-Cy5.5
PSCA CAR-Vδ1 T
Mock Vδ1 T
TCR Vδ1+Granzyme+B (%)
**
TCR Vδ1+IFN-γ+ (%)
***
F
Normalized cell index
ns
ns
ns
PSCA CAR-Vδ1 T
No antibody
NKp30 antibody
NKG2D antibody
DNAM-1 antibody
Tumor only
G
Tumor cells
Activating ligands
PSCA
IFN-γ
Killing
Killing
PSCA CAR
Granzyme B
TCR Vδ1
NK receptors
NKG2D
DNAM-1
NKp30
PSCA CAR-Vδ1 T cells

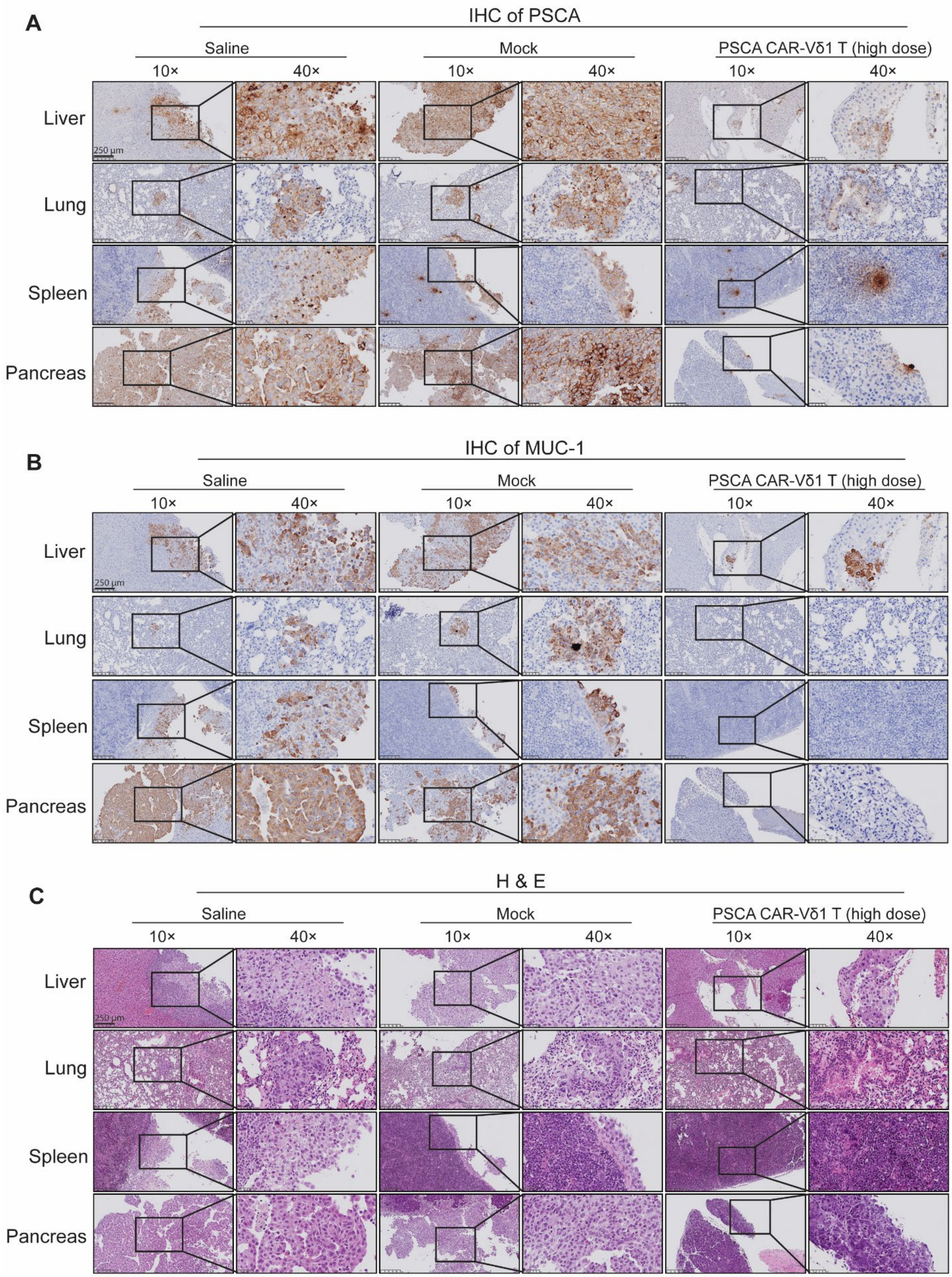
Supplementary Figure 2. PSCA CAR-Vδ1 T cells retain functional properties after a freeze-thaw cycle (related to Figure 2).

(A) PSCA CAR transduction efficiency tested by FACS with PE conjugated anti-EGFR antibody (n = 6). **(B)** Cytotoxicity comparison between fresh and after frozen PSCA-directed CAR Vδ1 T cells against $PSCA^{high}$ Capan-1 cells (n = 4). **(C)** Cell index comparison after 12 hours coculture from (B). Data shown are the mean ± SEM. **(D)** Serial killing assay was performed using a real time cytotoxicity assay with an E:T ratio of 1:2 (5,000 PSCA CAR-Vδ1 T effector cells to 10,000 tumor cells added at each of the three indicated intervals) and presented as the growth index of the residual Capan-1 cells. A total of 5,000 PSCA CAR-Vδ1 T and 30,000 tumor cells were added over the course of the experiment (n = 6, duplicate wells). **(E)** Granzyme B and IFN-γ expression and summary data by PSCA CAR-Vδ1 T cells compared to mock transduced Vδ1 T cells, after coculture with $PSCA^{high}$ Capan-1 cells with E: T of 1:1 (n = 4 for Granzyme B, n = 6 for IFN-γ); $**P < 0.01$, $***P < 0.001$ (t test). **(F)** *In vitro* activation receptor blocking assay. The concentration of each receptor blocking antibody is 2 μg/mL. ns, not significant, $*P < 0.05$ (one-way ANOVA). **(G)** Diagram showing the tumor-targeting triple mechanisms of PSCA Vδ1 T cells, mediated by NK activating receptors, Vδ1 TCR, and PSCA CAR.


A
PBMCs
γδT cells enrich & sort
αβT cells culure & sort
Vδ1 T cells
Vδ2 T cells
αβ T cells
Retro/PSCA-tEGFR
CAR transduction
in vitro expansion
1~2 weeks
CAR-Vδ1 T cells
CAR-Vδ2 T cells
CAR-αβT cells
B
TCR αβ-FITC
TCR Vδ1-FITC
TCR Vδ2-FITC
Percentage of TCR (%)
αβT cells
Vδ1T cells
Vδ2T cells
C
CAR-Vδ1 T cells
CAR-Vδ2 T cells
CAR-αβT cells
DAPI
Annexin-V-PE
D
TCR αβ-FITC
TCR-Vδ1-FITC
TCR-Vδ2-FITC
tEGFR-APC
E
Capan-1
Vδ1 T, Vδ2 T, αβ T
with/out PSCA CAR
Mix
Culture Dish
In vitro effector cells &
tumor cells co-culture
F
Capan-1
αβ T cells
Vδ1 T cells
Vδ2 T cells
Cytotoxicity (%)
G
Capan-1
CAR αβ T cells
CAR Vδ1 T cells
CAR Vδ2 T cells
Cytotoxicity (%)

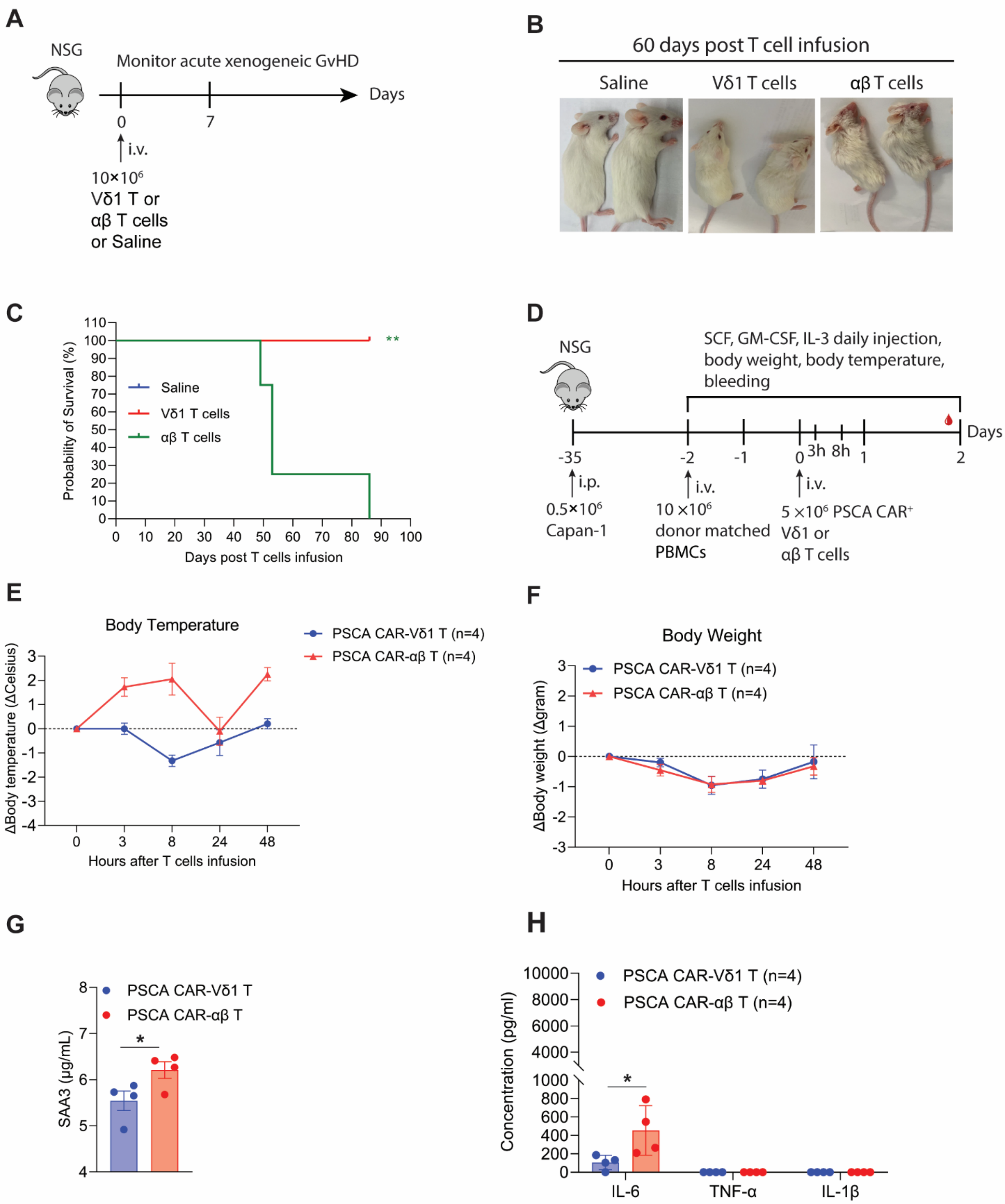
Supplementary Figure 3. Vδ1, Vδ2, and αβ T cells and PSCA CAR-Vδ1, PSCA CAR-Vδ2, and PSCA CAR-αβ T cells showed similar antitumor efficacy *in vitro* (related to Figure 2).

(A) Scheme shows the Vδ1, Vδ2, and αβ T cells and the generation process for their PSCA CAR engineered derivatives. **(B)** Representative FACS histogram shows the cell purity of αβ T cells (top), Vδ1 T cells (middle), and Vδ2 T cells (bottom) by measuring the TCR-αβ TCR-Vδ1, and TCR-Vδ2, respectively (left panel). Summary of (B) (right panel). Data shown are the mean ± SEM. **(C)** Representative FACS plot show the Vδ1 T cells, Vδ2 T cells, and αβ T cells viability manifested by DAPI and Annexin-V staining. **(D)** Representative FACS plot shows the comparable PSCA CAR expression level on αβ T cells, Vδ1 T cells, and Vδ2 T cells before *in vitro* antitumor efficacy assessment. **(E)** *In vitro* cytotoxicity experiment design for (F) and (G). **(F)** Cytotoxicity comparison between Vδ1 T cells, Vδ2 T cells, and αβ T cells against Capna-1 cells. Data shown are the mean ± SEM. **(G)** Cytotoxicity comparison between PSCA CAR-Vδ1 T cells, PSCA CAR-Vδ2 T cells, and PSCA CAR-αβ T cells against Capan-1 cells. Data shown are the mean ± SEM.


A
IHC of PSCA
Saline
Mock
PSCA CAR-Vδ1 T (high dose)
10×
40×
Liver
Lung
Spleen
Pancreas
250 μm
B
IHC of MUC-1
Saline
Mock
PSCA CAR-Vδ1 T (high dose)
10×
40×
Liver
Lung
Spleen
Pancreas
250 μm
C
H & E
Saline
Mock
PSCA CAR-Vδ1 T (high dose)
10×
40×
Liver
Lung
Spleen
Pancreas
250 μm

Supplementary Figure 4. Absence of tumor cells after PSCA CAR-Vδ1 T cell treatment (related to Figure 3).

Representative images from **(A)** anti-human PSCA immunohistochemistry (IHC), **(B)** anti-human MUC1 IHC, and **(C)** H&E staining of the indicated tissues from the metastatic PC mouse model after the indicated treatments. Scale bar, 250 μm.



A
NSG
Monitor acute xenogeneic GvHD
Days
0
7
i.v.
10×10⁶
Vδ1 T or
αβ T cells
or Saline
B
60 days post T cell infusion
Saline
Vδ1 T cells
αβ T cells
C
Probability of Survival (%)
Saline
Vδ1 T cells
αβ T cells
**
Days post T cells infusion
D
NSG
SCF, GM-CSF, IL-3 daily injection,
body weight, body temperature,
bleeding
Days
-35
-2
-1
0
3h
8h
1
2
i.p.
0.5×10⁶
Capan-1
i.v.
10 ×10⁶
donor matched
PBMCs
i.v.
5 ×10⁶ PSCA CAR⁺
Vδ1 or
αβ T cells
E
Body Temperature
PSCA CAR-Vδ1 T (n=4)
PSCA CAR-αβ T (n=4)
ΔBody temperature (ΔCelsius)
Hours after T cells infusion
F
Body Weight
PSCA CAR-Vδ1 T (n=4)
PSCA CAR-αβ T (n=4)
ΔBody weight (Δgram)
Hours after T cells infusion
G
PSCA CAR-Vδ1 T
PSCA CAR-αβ T
SAA3 (μg/mL)
*
H
PSCA CAR-Vδ1 T (n=4)
PSCA CAR-αβ T (n=4)
Concentration (pg/ml)
*
IL-6
TNF-α
IL-1β

Supplementary Figure 5. PSCA CAR-Vδ1 T cells induce less GvHD and CRS compared to αβ T cells (related to Figure 6).

(A) Experimental design of the GvH response of Vδ1 T cells using an NSG mouse xenograft model. Donor-matched αβ T cells were included as a control. **(B)** Representative mice images show the evidence of GvHD only in mice that received αβ T cells. **(C)** Kaplan-Meier survival curve of experimental mice over time (n = 4), $^{**}P < 0.01$ [log-rank (Mantel-Cox) test]. **(D)** Schema of experimental design for CRS study. **(e)** Body temperature (Celsius) change was monitored after T cell infusion. **(F)** Body weight (gram) change was monitored after T cell infusion. **(G)** Mouse serum amyloid A3 protein was measured by using the Mouse SAA-3 ELISA kit according to the manufacturer instructions (Sigma-Aldrich, EZMSAA3-12K). **(H)** CRS related cytokine hIL-6, TNF-α, hIL-1β were measured in serum by using the cytometric beads array (BD, 551809, 558279) at 48 hours after T cells infusion. For (E-H) n = 4 for each group of mice and error bars indicate SEM; for (G) and (H) $^{*}P < 0.05$ (t test).